Weekly Engineering Assignment 3 - Group 3 - S23

Driveway Cost Table

My boss has requested a table that displays all the prices associated with different widths and lengths.
So, I will make the table through python, the table will be intuitive, and it will look good.

Section 1: Table Creation

1. Build a table incrementing the lengths for width = 10 (that is, the first column of costs in the example)

To create the price table for fixed widths and lengths ranging from 10 to 30 feet, we can start by using a for loop to generate a list of lengths. We can do this using the range function to create a sequence of integers from 10 to 30. Next, for each length, we can calculate the number of concrete boxes needed, assuming a fixed width of 10 feet. Calculate the volume of the driveway, divide it into volume of one single bag, and multiply it by 1.1 (spillage).

In [11]:
```
# fixed tickness 4inch
length_lst = [i for i in range (10, 31)]
area_lst = [x * 10 for x in length_lst] # fixed width
need_bag_lst = [round(y / 2 * 1.1) for y in area_lst] # multiply by 1.1(calculate spil
price_lst = [round(z * 5.04) for z in need_bag_lst]
# length(10 ~ 30) * 10(width feet) / 2(square feet) * 1.1 * 5.04(price for each bag) =
print("\nThis sample illustrates that the width is a fixed value of 10.")
print("length =", end = "")
for i in length_lst:
    print(f"{i:4}", end = " ")
print("")
print("price = ", end = "")
for j in price_lst:
    print(f"{j:4}", end = " ")
```

```
This sample illustrates that the width is a fixed value of 10.
length =  10   11   12   13   14   15   16   17   18   19   20   21   22   23   24
25   26   27   28   29   30
price =  277  307  333  363  388  413  444  474  499  529  554  585  610  640  665  6
96  721  746  776  806  832
```

2. Vary the width from 10 to 24 feet in increments of 2 feet

Then put your code into the code cell below. Don't forget to print your results. For making the table looks like example above, firstly, print out the width for first colum. And the first row will be length. Therefore we can calculate the price when I print put the each length.

In [12]:
```
def calculate_price(length, width): # thickness 4 inch , one bag concrete = 4 inch * 2
    volume = length * width
    need_bags = round(volume / 2 * 1.1) # don't have to calculate the 4inch's tickness
    price = round(need_bags * 5.04)
    return price

width_lst = [i for i in range (10 , 25 ,2)]
length_lst = [i for i in range (10, 31)]
print("      ", end = "") #I just added a space to make it look good.
for i in width_lst:
    print(f"{i:4}", end = " ")
print("")
for x in length_lst:
    print(x, end = "    ")
    for i in range (0,8):
        price = calculate_price(x, width_lst[i])
        print(f"{price:4}", end = " ")
    print("")
```

```
        10   12   14   16   18   20   22   24
10     277  333  388  444  499  554  610  665
11     307  368  428  489  549  610  670  731
12     333  398  464  534  600  665  731  796
13     363  433  504  575  650  721  791  867
14     388  464  544  620  701  776  852  932
15     413  499  585  665  746  832  917  998
16     444  534  620  711  796  887  978 1063
17     474  564  660  756  847  942 1038 1129
18     499  600  701  796  897  998 1099 1200
19     529  630  736  842  948 1053 1159 1265
20     554  665  776  887  998 1109 1220 1331
21     585  701  816  932 1048 1164 1280 1396
22     610  731  852  978 1099 1220 1341 1462
23     640  766  892 1018 1149 1275 1401 1532
24     665  796  932 1063 1200 1331 1462 1598
25     696  832  973 1109 1250 1386 1522 1663
26     721  867 1008 1154 1295 1441 1588 1729
27     746  897 1048 1200 1346 1497 1648 1794
28     776  932 1089 1240 1396 1552 1709 1865
29     806  963 1124 1285 1446 1608 1769 1930
30     832  998 1164 1331 1497 1663 1830 1996
```

3. Enter the data into a data structure

If we need only the price data, can make it easily by append method instead of print(section 2).

In [10]:
```
data_lst = [[] for _ in range (21)] # 10 - 30 each length will be filled to empty list

for x in range (21):
    for i in range (0,8):
        price = calculate_price(length_lst[x], width_lst[i])
        data_lst[x].append(price)

for y in data_lst:
    print(y)
```

```
[277, 333, 388, 444, 499, 554, 610, 665]
[307, 368, 428, 489, 549, 610, 670, 731]
[333, 398, 464, 534, 600, 665, 731, 796]
[363, 433, 504, 575, 650, 721, 791, 867]
[388, 464, 544, 620, 701, 776, 852, 932]
[413, 499, 585, 665, 746, 832, 917, 998]
[444, 534, 620, 711, 796, 887, 978, 1063]
[474, 564, 660, 756, 847, 942, 1038, 1129]
[499, 600, 701, 796, 897, 998, 1099, 1200]
[529, 630, 736, 842, 948, 1053, 1159, 1265]
[554, 665, 776, 887, 998, 1109, 1220, 1331]
[585, 701, 816, 932, 1048, 1164, 1280, 1396]
[610, 731, 852, 978, 1099, 1220, 1341, 1462]
[640, 766, 892, 1018, 1149, 1275, 1401, 1532]
[665, 796, 932, 1063, 1200, 1331, 1462, 1598]
[696, 832, 973, 1109, 1250, 1386, 1522, 1663]
[721, 867, 1008, 1154, 1295, 1441, 1588, 1729]
[746, 897, 1048, 1200, 1346, 1497, 1648, 1794]
[776, 932, 1089, 1240, 1396, 1552, 1709, 1865]
[806, 963, 1124, 1285, 1446, 1608, 1769, 1930]
[832, 998, 1164, 1331, 1497, 1663, 1830, 1996]
```

Section 2: Table Search with Fixed Budget

Assume three situations. use, if elif else for 3 situations

first 'if' statement will find the situation when the budget is not enough to build smalles driveway(10feet length, 10feet width)

Second 'elif' statement will find the situation when the budget is enough to build largest drvieway(30feet length, 10feet width)

Lastly, 'else' statement will find the the driveway with the best length.

All of the step will calculate the (required budget, remain budget)

In [14]:

```
# In this cell, test the case with a budget too small to afford any driveway
budget = float(input("Enter your budget"))
print("The budget is $%.2f" %budget)
if budget < data_lst[0][0]: #data_lst[0][0] smallest driveway
    print("Your budget is too small, to bulid smallest driveway(10 feet wide).")
    print("Your minimum required budget is: $%d" %(data_lst[0][0] - budget))

elif budget > data_lst[-1][0]:
    print("Your budget is enough to build biggest driveway(10 feet wide and 30 feet le
    print("Your remaining budget is: $%d" %(budget - data_lst[-1][0]))

else:
    for i in range (0, 21):
        if budget < data_lst[i][0]:
            idx = i-1 # find the idx, appropriate one
            print("The longest driveway you can create on a budget is as follows:", er
            print((10 + (i-1)), end = "feet\n") # calculate the length by using the ic
            print("Your remaining budget is: $%d" %(budget - data_lst[i-1][0]))
            break
```

```
The budget is $250.00
Your budget is too small, to bulid smallest driveway(10 feet wide).
Your minimum required budget is: $27
```

In [15]:

```
# In this cell, test the case with a budget large enough to afford any size driveway
budget = float(input("Enter your budget"))
print("The budget is $%.2f" %budget)
if budget < data_lst[0][0]: #data_lst[0][0] smallest driveway
    print("Your budget is too small, to bulid smallest driveway(10 feet wide).")
    print("Your minimum required budget is: $%d" %(data_lst[0][0] - budget))

elif budget > data_lst[-1][0]:
    print("Your budget is enough to build biggest driveway(10 feet wide and 30 feet le
    print("Your remaining budget is: $%d" %(budget - data_lst[-1][0]))

else:
    for i in range (0, 21):
        if budget < data_lst[i][0]:
            idx = i-1 # find the idx, appropriate one
            print("The longest driveway you can create on a budget is as follows:", er
            print((10 + (i-1)), end = "feet\n") # calculate the length by using the ic
            print("Your remaining budget is: $%d" %(budget - data_lst[i-1][0]))
            break
```

```
The budget is $850.00
Your budget is enough to build biggest driveway(10 feet wide and 30 feet length).
Your remaining budget is: $18
```

In [16]:

```
# In this cell, test the case with a budget that can only afford a driveway somewhere
budget = float(input("Enter your budget"))
print("The budget is $%.2f" %budget)
if budget < data_lst[0][0]: #data_lst[0][0] smallest driveway
    print("Your budget is too small, to bulid smallest driveway(10 feet wide).")
    print("Your minimum required budget is: $%d" %(data_lst[0][0] - budget))

elif budget > data_lst[-1][0]:
    print("Your budget is enough to build biggest driveway(10 feet wide and 30 feet le
    print("Your remaining budget is: $%d" %(budget - data_lst[-1][0]))

else:
    for i in range (0, 21):
        if budget < data_lst[i][0]:
            idx = i-1 # find the idx, appropriate one
            print("The longest driveway you can create on a budget is as follows:", er
            print((10 + (i-1)), end = "feet\n") # calculate the length by using the ic
            print("Your remaining budget is: $%d" %(budget - data_lst[i-1][0]))
            break
```

```
The budget is $450.00
The longest driveway you can create on a budget is as follows: 16feet
Your remaining budget is: $6
```

Section 3: Automatic Table Creation

First function - Cost return, required for 1 square foot * thickness

Secound function - Price table output function according to arbitrary length and width

Third function - section2 -> make it function

In [69]:
```
def driveway_cost(thickness, cost, volume): # cost = $5.04, each_bag can fill 1 square
    need_bag = thickness / volume * 1.1 # consider the spillage.
    need_price = need_bag * cost
    return need_price

def driveway_table(lengths_range, width_range, concrete_price):
    width_lst = [i for i in range (width_range[0], width_range[1] ,width_range[2])]
    length_lst = [i for i in range (lengths_range[0], lengths_range[1], lengths_range[
    print("     ", end = "") #I just added a space to make it look good.
    for i in width_lst:
        print(f"{i:4}", end = " ")
    print("")
    for x in length_lst:
        print(x, end = "   ")
        for i in range (0, len(width_lst)):
            price = round(x * width_lst[i] * concrete_price)
            print(f"{price:4}", end = " ")
        print("")

each_bag_volume = 2 * 4 # 2 square foot * 4 inch fixed value
print("the thickness of the road to be constructed(inch): ")
thickness = float(input())
print(thickness)
concrete_bag_price = 5.04 # fixed value of price

price = driveway_cost(thickness, concrete_bag_price, each_bag_volume) # 10ft3
```

```
the thickness of the road to be constructed(inch):
8.0
```

In [78]:
```
#1 a single driveway
print("Your driveways thickness: %.2f feet" %(thickness/12))
print("column = length, row = width")
lengths_range = (10, 11, 1)
width_range = (10, 11, 1)
driveway_table(lengths_range, width_range, price)
```

```
Your driveways thickness: 0.67 feet
column = length, row = width
       10
10    554
```

In [76]:
```
#2 A table with lengths from 10 to 30, inclusive, and widths from 10 to 24, inclusive,
print("Your driveways thickness: %.2f feet" %(thickness / 12))
print("column = length, row = width")
lengths_range = (10, 16, 1)
width_range = (10, 16, 1)
driveway_table(lengths_range, width_range, price)
```

```
Your driveways thickness: 0.67 feet
column = length, row = width
       10   11   12   13   14   15
10    554  610  665  721  776  832
11    610  671  732  793  854  915
12    665  732  798  865  931  998
13    721  793  865  937 1009 1081
14    776  854  931 1009 1087 1164
15    832  915  998 1081 1164 1247
```

In [77]:

```
#3 5 x 5 table
print("Your driveways thickness: %.2f feet" %(thickness / 12))
print("column = length, row = width")
lengths_range = (10, 31, 1)
width_range = (10, 25, 2)
driveway_table(lengths_range, width_range, price)
```

```
Your driveways thickness: 0.67 feet
column = length, row = width
       10   12   14   16   18   20   22   24
10    554  665  776  887  998 1109 1220 1331
11    610  732  854  976 1098 1220 1342 1464
12    665  798  931 1064 1198 1331 1464 1597
13    721  865 1009 1153 1297 1441 1586 1730
14    776  931 1087 1242 1397 1552 1708 1863
15    832  998 1164 1331 1497 1663 1830 1996
16    887 1064 1242 1419 1597 1774 1951 2129
17    942 1131 1319 1508 1696 1885 2073 2262
18    998 1198 1397 1597 1796 1996 2195 2395
19   1053 1264 1475 1685 1896 2107 2317 2528
20   1109 1331 1552 1774 1996 2218 2439 2661
21   1164 1397 1630 1863 2096 2328 2561 2794
22   1220 1464 1708 1951 2195 2439 2683 2927
23   1275 1530 1785 2040 2295 2550 2805 3060
24   1331 1597 1863 2129 2395 2661 2927 3193
25   1386 1663 1940 2218 2495 2772 3049 3326
26   1441 1730 2018 2306 2595 2883 3171 3459
27   1497 1796 2096 2395 2694 2994 3293 3593
28   1552 1863 2173 2484 2794 3105 3415 3726
29   1608 1929 2251 2572 2894 3216 3537 3859
30   1663 1996 2328 2661 2994 3326 3659 3992
```

In [86]:

```
#4 20 by 4 table
print("Your driveways thickness: %.2f feet" %(thickness / 12))
print("column = length, row = width")
lengths_range = (10, 30, 1)
width_range = (5, 9, 1)
driveway_table(lengths_range, width_range, price)
```

```
Your driveways thickness: 0.67 feet
column = length, row = width
        5    6    7    8
10    277  333  388  444
11    305  366  427  488
12    333  399  466  532
13    360  432  505  577
14    388  466  543  621
15    416  499  582  665
16    444  532  621  710
17    471  565  660  754
18    499  599  699  798
19    527  632  737  843
20    554  665  776  887
21    582  699  815  931
22    610  732  854  976
23    638  765  893 1020
24    665  798  931 1064
25    693  832  970 1109
26    721  865 1009 1153
27    748  898 1048 1198
28    776  931 1087 1242
29    804  965 1125 1286
```

In [79]:

```python
def check_budget(table, budget):
    print("The budget is $%.2f" %budget)
    if budget < table[0][0]: #data_lst[0][0] smallest driveway
        print("Your budget is too small, to bulid smallest driveway(10 feet wide).")
        print("Your minimum required budget is: $%d" %(table[0][0] - budget))

    elif budget > table[-1][0]:
        print("Your budget is enough to build biggest driveway(10 feet wide and 30 fee
        print("Your remaining budget is: $%d" %(budget - table[-1][0]))

    else:
        for i in range (0, 21):
            if budget < table[i][0]:
                idx = i-1 # find the idx, appropriate one
                print("The longest driveway you can create on a budget is as follows:"
                print((10 + (i-1)), end = "feet\n") # calculate the length by using th
                print("Your remaining budget is: $%d" %(budget - table[i-1][0]))
                break
```

In [82]:

```python
budget = int(input(print("Enter your budget: ", end = "")))
print(budget)
check_budget(data_lst, budget)
```

```
Enter your budget: 250
The budget is $250.00
Your budget is too small, to bulid smallest driveway(10 feet wide).
Your minimum required budget is: $27
```

In [83]:

```python
budget = int(input(print("Enter your budget: ", end = "")))
print(budget)
check_budget(data_lst, budget)
```

```
Enter your budget: 500
The budget is $500.00
The longest driveway you can create on a budget is as follows: 18feet
Your remaining budget is: $1
```

In [84]:
```
budget = int(input(print("Enter your budget: ", end = "")))
print(budget)
check_budget(data_lst, budget)
```

```
Enter your budget: 1000
The budget is $1000.00
Your budget is enough to build biggest driveway(10 feet wide and 30 feet length).
Your remaining budget is: $168
```

we can get a well-organized table by setting the thickness, length, and width. It looks good and tells you the longest driveway according to your budget input. If you want, you can save the data set that you used once in a list in list structure variable.